

ELAT, Israel

WMO: 401990

Lat: 29.5526N

Lon: 34.9520E

Elev: 22

StdP: 101.06

Time Zone: 2.00 (ISR)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 8.6	(c) 9.9	(d) -7.6	(e) 2.0	(f) 21.1	(g) -5.1	(h) 2.5	(i) 21.8	(j) 9.9	(k) 18.8	(l) 9.2	(m) 18.7	(n) 2.5	(o) 30	(p) 0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	42.8	21.1	41.7	20.8	40.6	20.7	24.2	35.7	23.5	35.3	22.9	35.1	5.8	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	15.2	31.1	19.6	14.3	30.8	18.8	13.7	30.5	73.1	35.9	70.3	35.2	68.1	35.0	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.2	(b) 9.4	(c) 8.7	(e) 5.9	(f) 45.3	(g) 1.2	(h) 1.2	(i) 5.0	(j) 46.2	(k) 4.3	(l) 46.9	(m) 3.6	(n) 47.6	(o) 2.8	(p) 48.5		
(a) 10.2	(b) 9.4	(c) 8.7	(e) 2.7	(f) 26.1	(g) 1.4	(h) 0.8	(i) 1.7	(j) 26.7	(k) 0.8	(l) 27.2	(m) 0.0	(n) 27.6	(o) -1.0	(p) 28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.9	16.1	17.9	21.1	25.0	29.5	32.5	34.2	34.1	31.7	27.8	22.5	17.7		
	DBStd	6.87	2.22	2.83	3.14	3.28	2.92	2.15	1.55	1.49	1.77	2.59	2.79	2.39		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	166	75	38	8	1	0	0	0	0	0	0	3	42		
	CDD10.0	5792	189	221	345	451	603	674	749	747	651	552	374	237		
	CDD18.3	2918	6	26	95	201	345	424	491	489	401	293	127	21		
	CDH23.3	41560	43	198	834	2214	4785	6714	8162	8081	5986	3453	969	121		
(13)	CDH26.7	25642	3	42	274	1032	2821	4444	5694	5609	3701	1711	292	18		
(14)	Wind	WSAvg	4.2	3.4	3.6	4.0	4.2	4.4	4.3	4.7	5.0	4.4	3.9	3.6		
(15) Precipitation	PrecAvg	24	3	4	3	3	1	0	0	0	0	3	3	5		
	PrecMax	62	23	67	22	28	13	0	0	0	0	35	27	43		
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0		
	PrecStd	16	5	10	5	6	3	0	0	0	0	8	6	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.8	30.2	34.2	38.2	42.2	44.0	44.1	43.8	41.8	39.0	33.8	29.1		
		MCWB	14.3	15.6	16.7	17.4	19.7	20.4	21.5	22.5	21.5	19.9	18.3	15.6		
	2%	DB	24.3	27.9	31.9	36.0	39.9	42.1	42.8	42.1	39.9	36.8	31.4	26.1		
		MCWB	13.8	14.5	16.0	17.2	19.3	20.1	21.3	21.5	21.5	20.2	17.5	14.9		
	5%	DB	23.0	25.8	29.8	34.0	38.0	40.8	41.4	41.0	38.7	35.0	29.9	24.3		
		MCWB	13.4	14.1	15.5	16.7	18.7	19.9	20.8	21.2	21.3	20.0	17.2	14.4		
	10%	DB	21.5	23.9	27.8	32.1	36.2	39.2	40.2	40.0	37.2	33.3	28.1	23.0		
		MCWB	12.7	13.7	15.0	16.3	18.2	19.8	20.7	21.4	21.2	19.8	16.6	13.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.8	19.2	20.0	20.6	22.0	23.1	24.9	25.5	25.0	23.5	20.7	19.4		
		MCDB	22.9	25.1	26.5	30.0	33.3	35.7	36.4	37.3	35.9	32.4	28.6	24.8		
	2%	WB	15.2	16.6	18.1	19.1	20.9	22.2	23.7	24.3	23.9	22.6	19.3	16.4		
		MCDB	22.4	24.7	27.4	30.5	34.5	35.8	36.3	36.2	35.1	32.2	28.6	24.2		
	5%	WB	13.9	15.1	16.8	18.1	20.1	21.4	23.0	23.6	23.1	21.7	18.3	15.1		
		MCDB	21.7	23.9	27.2	30.8	34.4	35.8	36.3	35.7	34.3	31.1	27.5	22.6		
	10%	WB	13.0	14.0	15.7	17.2	19.3	20.8	22.3	23.0	22.5	21.0	17.5	14.1		
		MCDB	20.4	22.9	26.4	30.2	33.7	35.7	35.8	35.4	33.7	30.6	26.4	21.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	11.3	11.5	12.3	12.9	13.5	13.0	12.4	11.8	11.1	11.1	10.9		
		MCDBR	12.5	13.8	13.9	14.2	14.4	14.9	13.9	13.3	12.9	12.5	12.4	12.4		
		MCWBR	6.2	6.2	6.0	5.4	5.0	4.4	4.2	4.0	4.2	4.7	5.4	6.0		
	5% WB	MCDBR	11.1	11.9	12.1	12.3	12.9	13.4	12.6	11.9	11.7	11.0	11.0	10.9		
		MCWBR	6.2	6.4	6.4	5.6	5.3	4.6	4.7	4.3	4.4	4.7	5.3	6.1		
(40) Clear-Sky Solar Irradiance	taub		0.364	0.405	0.467	0.494	0.499	0.420	0.425	0.433	0.430	0.436	0.373	0.346		
	taud		2.251	2.108	1.913	1.876	1.892	2.152	2.168	2.153	2.148	2.110	2.272	2.340		
	Ebn at Noon		857	850	818	811	806	868	863	853	843	807	840	861		
	Edh at Noon		115	145	188	202	200	153	150	151	147	143	112	101		
(44) All-Sky Solar Radiation	RadAvg		3.83	4.79	6.15	7.16	7.87	8.52	8.23	7.65	6.68	5.36	4.25	3.62		
	RadStd		0.16	0.18	0.20	0.16	0.26	0.09	0.13	0.10	0.15	0.16	0.13	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.36	N/A	-1.26	N/A	N/A	N/A	N/A	N/A	+135	+146
(47)	Regional (11 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	+146

Nomenclature: See separate page